Local Complete-Graph Optimality at Fitness Two and Strong-Selection Rigidity under Death–Birth Updating

Alec Kriebel
Independent Researcher
me@aleckriebel.com
ORCID 0009-0001-9320-500X

August 21, 2026

Abstract

We study fixation from a uniformly random single mutant under death–Birth updating on finite loopless directed weighted population structures. After normalizing each target's incoming weights, the dynamics are parametrized by a loopless row-stochastic replacement kernel; let J_n denote the uniform complete kernel. We prove two complementary extremality results. First, at mutant fitness two, J_n is a strict nondegenerate local maximizer in the full normalized kernel polytope for every $n \geq 3$. This includes arbitrary directed and nonreversible perturbations. The proof represents the fixation committor as a completely alternating coverage function of a fair-geometric union dual, converts fixation into a stationary collision observable, and decomposes the inverse-mean Hessian into standard, symmetric-balanced, and antisymmetric-balanced permutation sectors with positive sector certificates valid for every population size, using exact rational computer assistance over an explicitly finite range. Second, we close the complete-support case left outside the prior strongly connected noncomplete-support obstruction. The leading strong-selection deficit is an explicit sum of squared discrepancies within incoming columns, with equality exactly at J_n after normalization. Together with the prior obstruction and a source-component bound, this shows that no fixed finite directed weighting is a strict amplifier for every beneficial fitness. We also prove global all-beneficial-fitness maximality on positive weighted triangles and within two symmetric weighted K_4 families. These results establish full local rigidity at fitness two and fixed-graph rigidity at strong selection, but not global complete-graph maximality at fitness two.

Keywords. death–Birth updating; fixation probability; evolutionary graph theory; coverage duality; local optimality; strong selection.

Mathematics Subject Classification (2020). 92D15; 60J10; 05C81.

Author summary

Spatial structure can make a beneficial mutation more or less likely to take over a finite population. For death–Birth updating, a natural unresolved question is whether the well-mixed population is globally extremal at fitness two. We prove two complementary pieces of that picture. At fitness two, the well-mixed kernel is strictly locally optimal against every infinitesimal perturbation of its normalized replacement kernel, including directed and nonreversible ones. In the strong-selection limit, every fixed finite structure is eventually suppressing unless its replacement dynamics are exactly well mixed. The local result comes from an exact ancestry process: fixation becomes a coverage probability, then a stationary collision rate, whose curvature splits into three symmetry types. The second result comes from a sum of squares measuring unequal competitors for the same death event. Weighted triangles provide a global bridge: every nonuniform positive triangle suppresses fixation at every beneficial fitness. The results do not show that the complete graph is globally best at fitness two or exclude graph sequences whose size threshold depends on fitness.

1 Introduction

The Moran process is a foundational finite-population model of selection and genetic drift (Moran, 1958). On a structured population, the fixation probability of a new mutant depends on both selection and the geometry of replacement. The complete graph is the well-mixed baseline, while directed and weighted evolutionary graphs can amplify or suppress selection under particular update rules and fitness ranges (Lieberman et al., 2005; Shakarian et al., 2012; Masuda and Ohtsuki, 2009; Allen et al., 2017, 2020). Birth–death and death–Birth updating behave especially differently (Kaveh et al., 2015): the latter exposes every mutant to a uniformly sampled death event before local competition, which severely constrains amplification (Hindersin and Traulsen, 2015; Tkadlec et al., 2020). Transient dB amplifiers have since been studied through exact computation and guided spectral edge deletion (Allen et al., 2020; Richter, 2023).

For a loopless directed weighting, multiply all incoming weights at a fixed target by the same positive number. This changes no death–Birth transition. The intrinsic object is therefore a loopless row-stochastic replacement kernel P, with a row indexed by the dead target and a column by a potential parent. Let J_n be the uniform off-diagonal kernel and write

$$D_n(P, r) = \rho_{\mathrm{dB}}(J_n, r) - \rho_{\mathrm{dB}}(P, r) \tag{1.1}$$

for the uniformly initialized fixation deficit. Global nonnegativity of $D_n(P, 2)$ for every n and every kernel induced by an undirected weighting remains open. This paper resolves two transverse rigidity problems for the same surface $D_n(P, r)$.

First, we approach the complete kernel in structure space while holding fitness fixed at two. For every $n \geq 3$ and every nonzero loopless row-zero perturbation δ,

$$D_n(J_n + \varepsilon\delta, 2) = \kappa_n(\delta)\varepsilon^2 + O(\varepsilon^3), \qquad \kappa_n(\delta) > 0. \tag{1.2}$$

Thus J_n is a strict nondegenerate local maximizer among all positive loopless row-stochastic replacement kernels. This is stronger than a reversible or undirected statement: column imbalance and directed circulation are both allowed. It is nevertheless genuinely local. Its neighborhood may depend on n, and it does not constrain a sequence that stays far from J_n. To our knowledge, no previous result establishes this strict local maximality in the full directed normalized kernel space.

Second, we hold an arbitrary finite structure fixed and let $r \to \infty$. On complete directed support,

$$D_n(P, r) = \frac{\mathcal{E}_{\mathrm{dir}}(P)}{n^2(n-2)r} + O(r^{-2}), \tag{1.3}$$

where $\mathcal{E}_{\mathrm{dir}}$ is an explicit incoming-column sum of squares. On the normalized kernel space its zero set is exactly $P = J_n$; at the level of unnormalized weights, this is the class dynamically equivalent to J_n. Theorem 1 of Tkadlec et al. (2020) applies to every fixed strongly connected noncomplete population structure, including directed and weighted supports. Thus complete directed support with unequal normalized weights is the residual strongly connected case. Equation (1.3) closes it sharply; together with the cited theorem and a source-component bound for non-strong supports, it rules out one fixed finite directed weighting that amplifies for every $r > 1$.

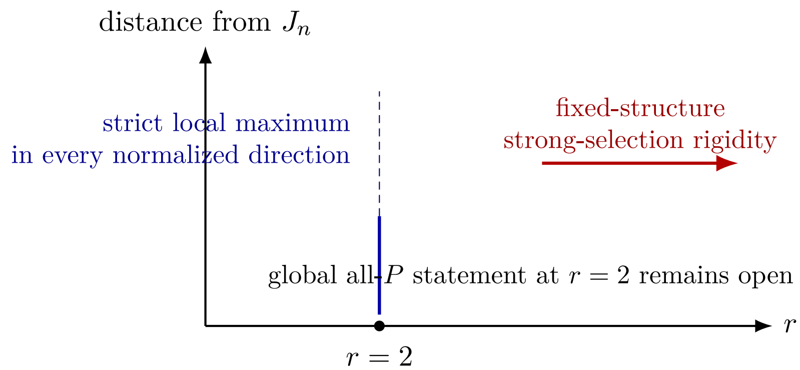

Figure 1: The two proved axes of the same complete-kernel deficit. The fitness-two theorem is local in the full normalized structure space and valid for every population size; the strong-selection theorem is global in fixed finite structure and asymptotic in fitness.

The mechanism behind (1.2) is different from ordinary weak-selection perturbation. Branching and coalescing genealogical duals under selection have a substantial population-genetic history (Krone and Neuhauser, 1997; Etheridge and Griffiths, 2009; Mano, 2009); related spatial dB work constructs the process and its dual on integer lattices (Foo et al., 2024). Here we specialize this framework to arbitrary finite directed kernels at $r = 2$: the update is an additive Boolean OR of a fair-geometric number of potential parents. Reversing the graphical maps produces a finite union-valued ancestry chain; no independent-lineage approximation is made. Its stationary law represents the fixation committor as a completely alternating coverage function, and a marked one-sample factorization turns the stationary mean ancestral-set size into a collision probability. At J_n, the first variation vanishes by rank projection. The Hessian is an S_n-invariant quadratic form on all loopless row-zero matrices. The tangent space splits without multiplicity into standard, symmetric-balanced, and antisymmetric-balanced sectors. We prove all three scalars positive, for every population size, by rank-recursion and phase-contraction certificates. The standard sector is exactly the column-imbalance direction and requires a separate signed three-channel argument.

The global low-order results sharpen the picture. Every nonuniform positive weighted triangle is a strict suppressor for every $r > 1$. Two maximally symmetric weighted K_4 families obey the same conclusion. These theorems are not inferred from local curvature; they follow from independent exact sum-of-squares certificates.

An earlier unrefereed research release established the strong-selection and low-order portions of this argument (Kriebel, 2026a). This manuscript incorporates and revises those proofs while adding the fitness-two theory. The following ledger makes that provenance explicit.

Component	Status relative to the earlier release
Complete-support strong-selection expansion and fixed-graph closure	Previously released; proof revised and integrated here.
Undirected support limit and support-degree consequence	Previously released; proof revised and integrated here.
Weighted triangles and the two symmetric K_4 families	Previously released; reorganized here.
$r = 2$ coverage representation and active collision identity	First included here.
Full directed Hessian decomposition and all-n positivity of its three sectors	First included here.
Comparison of the local and strong-selection quantifier axes	First included here.

The quantifiers matter. The fixed-graph theorem proves

$$\forall P\ \exists r_0(P)\ \forall r > r_0(P): \quad D_n(P, r) > 0$$

unless $P = J_n$. It does not rule out a single fitness-independent sequence (P_n) satisfying, for each fixed r, $D_n(P_n, r) < 0$ once n is sufficiently large. Such growing-family quantifiers are substantive: Svoboda et al. (2024) construct one family that is independent of fitness and whose sufficiently large members simultaneously amplify under both Bd and dB updating for every fixed $1 < r < 1.2$. We keep this distinction explicit throughout. Separate unrefereed companion work by the present author constructs a fitness-independent family whose sufficiently large members simultaneously amplify both rules for every fixed $1 < r < R_{\text{hyb}}$, where $R_{\text{hyb}} = 1.5028569127905696\ldots > 3/2$ (Kriebel, 2026b). That bounded-interval construction does not resolve the all-r growing-family question.

The paper is organized as follows. Section 2 states the model and theorem suite. Sections 3–4 prove the fitness-two local theorem. Section 5 treats fixed structures at strong selection. Section 6 gives the global low-order classifications. The appendices contain the exact phase certificates, the symmetric K_4 algebra, and a quantifier ledger.

2 Model and main results

Let $V = \{1, \ldots, n\}$ and let $w_{uv} \geq 0$ be the weight with which source u competes to replace target v. We assume $w_{vv} = 0$ and every incoming degree $d_v^- = \sum_{u \neq v} w_{uv}$ is positive. Normalize

$$P_{vu} = \frac{w_{uv}}{d_v^-}. \tag{2.1}$$

Write $P(W)$ for this kernel. Then $P(W)$ is loopless and row stochastic. Conversely, every such P is a normalized directed replacement structure. This quotient removes exactly the dynamically irrelevant positive rescalings of individual incoming columns. When weights rather than their normalized kernel are displayed, we use the shorthand $\rho_{\text{dB}}(W, r) := \rho_{\text{dB}}(P(W), r)$.

A uniformly chosen target dies. If the current mutant set is S and mutant fitness is $r > 0$, the probability that target v receives a mutant parent is

$$f_r(P_{vS}) = \frac{r P_{vS}}{1 + (r-1) P_{vS}}, \qquad P_{vS} = \sum_{u \in S} P_{vu}. \tag{2.2}$$

The all-resident and all-mutant states are absorbing. We write $\rho_{\mathrm{dB}}(P, r)$ for fixation probability averaged over a uniformly random singleton mutant. For the complete kernel

$$(J_n)_{vu} = \frac{1}{n-1} \quad (u \neq v), \tag{2.3}$$

one-dimensional first-step equations give

$$\rho_{\mathrm{dB}}(J_n, r) = \frac{n-1}{n} \frac{1 - r^{-1}}{1 - r^{-(n-1)}}. \tag{2.4}$$

At $r = 1$, the right-hand side is understood by continuity and equals $1/n$.

Let

$$\mathcal{T}_n = \{\delta \in \mathbb{R}^{n \times n} : \delta_{ii} = 0,\ \delta \mathbf{1} = 0\} \tag{2.5}$$

be the tangent space to the normalized kernel polytope at J_n, and let

$$\mathcal{B}_n = \{\delta \in \mathcal{T}_n : \mathbf{1}^{\mathsf{T}} \delta = 0\} \tag{2.5b}$$

be its balanced, or bistochastic, tangent subspace.

Theorem 1 (fitness-two strict local optimality)**.** *For every $n \geq 3$ there is a quadratic form $\mathcal{R}_n^{(2)}$ on $\mathcal{T}_n$ such that, for every $\delta \in \mathcal{T}_n$ and all sufficiently small ε for which $J_n + \varepsilon\delta$ is nonnegative,*

$$\frac{1}{n\rho_{\mathrm{dB}}(J_n + \varepsilon\delta, 2)} = \frac{1}{m_n} + \varepsilon^2 \mathcal{R}_n^{(2)}(\delta) + O(\varepsilon^3), \tag{2.6}$$

where

$$m_n = \frac{(n-1)2^{n-2}}{2^{n-1} - 1}. \tag{2.7}$$

The quadratic form $\mathcal{R}_n^{(2)}$ is positive definite on $\mathcal{T}_n$. Consequently, the first variation vanishes for every $\delta \in \mathcal{T}_n$, and, for every nonzero $\delta \in \mathcal{T}_n$,

$$\left.\frac{d}{d\varepsilon} \rho_{\mathrm{dB}}(J_n + \varepsilon\delta, 2)\right|_0 = 0, \qquad \left.\frac{d^2}{d\varepsilon^2} \rho_{\mathrm{dB}}(J_n + \varepsilon\delta, 2)\right|_0 = -\frac{2m_n^2}{n} \mathcal{R}_n^{(2)}(\delta) < 0. \tag{2.8}$$

Thus J_n is a strict nondegenerate local maximizer in the full positive loopless row-stochastic kernel polytope. For n = 2 *every positive weighting induces the same two-vertex chain and ties the baseline.*

For clarity, the positive-kernel hypothesis used by the duality proof loses no tangent direction. For fixed nonzero $\delta \in \mathcal{T}_n$, put $M = \max_{i \neq j} |\delta_{ij}|$. Whenever $|\varepsilon| < 1/[(n-1)M]$, every off-diagonal entry of $J_n + \varepsilon\delta$ is strictly positive. The zero direction is trivial. Thus the expansion in Theorem 1 is proved entirely inside the positive kernel polytope before being interpreted at its tangent space.

For a positive normalized kernel on complete directed support, define the scale-free defect

$$\mathcal{E}_{\mathrm{dir}}(P) = \sum_{v} \sum_{\substack{u<z \\ u,z \neq v}} \frac{(P_{vu} - P_{vz})^2}{P_{vu} P_{vz}}. \tag{2.9}$$

For unnormalized weights W, set $\mathcal{E}_{\mathrm{dir}}(W) := \mathcal{E}_{\mathrm{dir}}(P(W))$; equivalently,

$$\mathcal{E}_{\mathrm{dir}}(W) = \sum_{v} \sum_{\substack{u<z \\ u,z \neq v}} \frac{(w_{uv} - w_{zv})^2}{w_{uv} w_{zv}}. \tag{2.9b}$$

Theorem 2 (complete-support strong-selection correction). *Let $n \geq 3$ and suppose $w_{uv} > 0$ for every $u \neq v$. For fixed W,*

$$\rho_{\mathrm{dB}}(J_n, r) - \rho_{\mathrm{dB}}(W, r) = \frac{\mathcal{E}_{\mathrm{dir}}(W)}{n^2(n-2)r} + O(r^{-2}) \qquad (r \to \infty). \tag{2.10}$$

The defect vanishes if and only if, for every target v, all incoming weights w_{uv} with $u \neq v$ are equal. In that equality class the entire chain, not merely the asymptotic expansion, equals the complete-graph chain.

Corollary 3 (no fixed finite universal dB amplifier). *No finite loopless directed weighting with positive incoming degrees is a strict dB amplifier for every $r > 1$. More precisely, it is strictly suppressing for all sufficiently large fitness unless it has complete support and is dynamically equivalent to J_n; in the latter case it ties the baseline for every fitness.*

For undirected weights, write s_i for the number of positive-weight neighbors of i.

Proposition 4 (undirected support limit). *For every finite connected undirected weighted graph,*

$$\lim_{r\to\infty} \rho_{\mathrm{dB}}(W, r) = \frac{1}{n}\sum_{i=1}^{n} \frac{s_i}{s_i + 1}. \tag{2.11}$$

Hence an incomplete support has the positive limiting deficit

$$\frac{n-1}{n} - \frac{1}{n}\sum_i \frac{s_i}{s_i+1} = \sum_i \frac{n-1-s_i}{n^2(s_i+1)}. \tag{2.12}$$

Finally, the complete kernel is globally extremal on the smallest nontrivial undirected order.

Theorem 5 (weighted triangles). *For every positive undirected weighted triangle and every $r > 1$,*

$$\rho_{\mathrm{dB}}(W, r) \leq \rho_{\mathrm{dB}}(J_3, r) = \frac{2r}{3(r+1)}. \tag{2.13}$$

Equality holds if and only if the three edge weights are equal.

The two symmetric K_4 families and their equality statements are recorded in Appendix B. They provide global four-vertex slices but are not needed for Theorems 1 or 2.

3 Boolean ancestry, coverage, and collision

This section is valid for every positive loopless row-stochastic P; reversibility is not used.

3.1 The fair-geometric union dual

At fitness two, (2.2) becomes $2x/(1+x)$. This is a finite additive graphical system in the sense of Harris (1978). Let L have the fair geometric law

$$\mathbb{P}(L = \ell) = 2^{-\ell}, \qquad \ell \geq 1, \tag{3.1}$$

and, conditionally on target v, draw $U_1, \ldots, U_L$ independently from row P_v. For any mutant set S,

$$\mathbb{P}(U_1, \ldots, U_L \notin S) = \mathbb{E}(1 - P_{vS})^L = \frac{1 - P_{vS}}{1 + P_{vS}}. \tag{3.2}$$

Thus the forward update at v is exactly the additive Boolean map

$$x_v \longleftarrow x_{U_1} \vee \cdots \vee x_{U_L}. \tag{3.3}$$

The transpose graphical map sends a nonempty ancestral set A to

$$A \longmapsto \begin{cases} (A \setminus \{v\}) \cup \{U_1, \dots, U_L\}, & v \in A, \\ A, & v \notin A. \end{cases} \tag{3.4}$$

Because $P_{vv} = 0$, the updated set in the first line omits v and is always nonempty; hence the dual remains a proper nonempty subset whenever it starts there.

Using the same graphical marks forward and backward gives the exact duality

$$\mathbb{P}_S(X_t \cap A \neq \varnothing) = \mathbb{P}_A(S \cap A_t \neq \varnothing). \tag{3.5}$$

Lemma 6 (ergodicity of the union dual)**.** *If every off-diagonal entry of P is positive, then the chain in (3.4) is irreducible and aperiodic on the proper nonempty subsets of V.*

Proof. From any set A of size at least two, choose a target $v \in A$, take $L = 1$, and sample $U_1 \in A \setminus \{v\}$. This removes v and changes no other element. Repetition reaches a singleton. Conversely, fix any proper nonempty destination B and choose $x \notin B$. A singleton can first be moved to $\{x\}$ using a one-sample update. An update targeted at x whose finite geometric burst visits every element of B and no element outside B then has output exactly B. Every event just described has positive probability because P is positive off the diagonal and every finite value of L has positive probability. Hence the chain is irreducible. Finally, every proper A has a target outside A; choosing it leaves A unchanged, so every state has a positive self-loop and the chain is aperiodic. □

Let Π_P denote its stationary law and put

$$m(P) = \mathbb{E}_{\Pi_P}|A|. \tag{3.6}$$

For a set function f and $i \notin S$, write

$$\Delta_i f(S) = f(S \cup \{i\}) - f(S), \qquad \Delta_T = \prod_{i \in T} \Delta_i \quad (T \cap S = \varnothing). \tag{3.6a}$$

The difference operators in the product commute.

Proposition 7 (coverage representation)**.** *The fixation committor at fitness two is*

$$h_P(S) = \mathbb{P}_{A \sim \Pi_P}(A \cap S \neq \varnothing), \tag{3.7}$$

and therefore

$$\rho_{\mathrm{dB}}(P, 2) = \frac{m(P)}{n}. \tag{3.8}$$

Moreover, for every nonempty T disjoint from S,

$$(-1)^{|T|+1} \Delta_T h_P(S) = \mathbb{P}_{\Pi_P}(A \cap S = \varnothing,\ T \subseteq A) \geq 0. \tag{3.9}$$

Thus h_P is a normalized completely alternating coverage function.

Proof. The finite forward chain absorbs almost surely. Indeed, from every nonabsorbing mutant set, a sequence of resident-target deaths followed by mutant-parent choices gives a positive-probability path to V; hence the finite chain has no other closed communicating class. Start the dual from V in (3.5). Its first active update sends V to $V \setminus \{v\}$ because all sampled parents differ from the target v; Lemma 6 then gives convergence to Π_P. Letting $t \to \infty$, the left side converges to the fixation probability from S, giving (3.7). Averaging $h_P(\{i\}) = \mathbb{P}(i \in A)$ over i proves (3.8). For fixed A, the indicator $\mathbf{1}_{\{A\cap S\neq\varnothing\}}$ has mixed difference $(-1)^{|T|+1}\Delta_T$ equal to $\mathbf{1}_{\{A\cap S=\varnothing,\ T\subseteq A\}}$; averaging proves (3.9). □

3.2 A one-sample active chain

The geometric burst can be factored into one-sample steps. Two phase spaces are required:

$$\mathcal{Z}_n = \{(C, v) : C \subseteq V \setminus \{v\}\}, \qquad \mathcal{Y}_n = \{(B, v) : \varnothing \neq B \subseteq V \setminus \{v\}\}. \tag{3.10}$$

The pre-sampling space $\mathcal{Z}_n$ permits the empty cache created when a singleton is stopped. Let $A : \mathcal{Z}_n \to \mathcal{Y}_n$ adjoin one source i sampled from row P_v. Let $R : \mathcal{Y}_n \to \mathcal{Z}_n$ be the fair continue-or-retarget channel: on continue it sends (B, v) to (B, v), while on stop it chooses w uniformly from B and sends (B, v) to $(B \setminus \{w\}, w)$. Row laws act on the right, so $K(P) = RA$ is a kernel on $\mathcal{Y}_n$ and $M(P) = AR$ is a kernel on $\mathcal{Z}_n$. Explicitly, from $y = (B, v)$, with $b = |B|$, $K(P)$ is the following fair mixture:

1. sample i from row P_v and move to $(B \cup \{i\}, v)$;
2. choose w uniformly from B, sample i from row P_w, and move to $((B \setminus \{w\}) \cup \{i\}, w)$.

Each branch has probability 1/2. The first branch continues a geometric burst; the second stops it and retargets the active ancestral mark.

Lemma 8 (active collision identity)**.** *The active kernel $K(P)$ is irreducible and aperiodic. Let $\nu(P)$ be its unique stationary law and put*

$$H(B, v) = \frac{1}{|B|}. \tag{3.11}$$

Then

$$\nu(P)H = \frac{1}{m(P)} = \frac{1}{n\rho_{\mathrm{dB}}(P, 2)}. \tag{3.12}$$

The map $P \mapsto K(P)$ is linear.

Proof. We include the marked-current calculation because it is the bridge from the coverage law to the Hessian. Let $T_v(A, B)$ be the transition kernel of one union-dual update with target fixed at v. Extend Π_P by zero off the proper nonempty subsets of V. For $v \notin C$, including $C = \varnothing$, define

$$\sigma_v(C) = \Pi_P(C \cup \{v\}), \qquad \eta_v(C) = \sum_A \Pi_P(A)T_v(A, C) - \Pi_P(C). \tag{3.13a}$$

The subtracted term is exactly the passive current from $A = C$, for which $v \notin A$ and no ancestral update occurs; hence $\eta_v(C)$ is the nonnegative active current arriving at output (C, v). Stationarity of the union dual, after summing its n equally likely target updates, gives

$$\sum_{v \notin B} \eta_v(B) = |B|\Pi_P(B). \tag{3.13}$$

If A_v adjoins one sample from row P_v, the fair-geometric resolvent identity gives

$$\lambda_v = \frac{\sigma_v + \eta_v}{2}, \qquad \eta_v = \lambda_v A_v. \tag{3.14}$$

For the rectangular channels just defined, after the sample, continuing contributes $\eta_v(B)/2$ to (B, v). Stopping, choosing w uniformly from B, and ending at $(B \setminus \{w\}, w)$ contributes, by (3.13), $\Pi_P(B)/2 = \sigma_w(B \setminus \{w\})/2$. Hence $\lambda AR = \lambda$. Thus λ is stationary on $\mathcal{Z}_n$ for $M = AR$. Since $\eta = \lambda A$, associativity gives $\eta RA = \eta$; hence η is stationary on $\mathcal{Y}_n$ for the active kernel $K = RA$. Both σ_v and η_v have total mass $\mathbb{P}_{\Pi_P}(v \in A)$, so

$$\sum_{v,C} \lambda_v(C) = m(P). \tag{3.15}$$

Normalizing gives the probability laws $\lambda/m(P)$ on $\mathcal{Z}_n$ and $\nu(P) = \eta/m(P)$ on $\mathcal{Y}_n$. The latter stationary law is unique: if $|B| \geq 2$, a stop at $w \in B$ followed by sampling a member of $B \setminus \{w\}$ shrinks B by one. From a singleton state, one or two positive stop moves reach any prescribed singleton–target pair $(\{j\}, v)$ with $j \neq v$. Positive continue moves then sample the remaining members of any prescribed active set. Hence $K(P)$ is irreducible; sampling an already present source on a continue move gives a self-loop, so it is also aperiodic. Dividing (3.13) by $|B|$ and summing over B yields

$$\nu(P)H = \sum_{k \geq 1} \frac{k\Pi_P(|A| = k)}{m(P)} \frac{1}{k} = \frac{1}{m(P)}. \tag{3.16}$$

Both active moves contain exactly one use of a row of P, proving linearity. $\square$

At the complete kernel, put $N = n - 1$. Direct substitution gives

$$\nu_0(B, v) = \frac{|B|}{nN2^{N-1}}, \qquad c_0 := \nu_0 H = \frac{2^N - 1}{N2^{N-1}} = \frac{1}{m_n}. \tag{3.17}$$

This agrees with (2.4) at $r = 2$.

The collision interpretation uses the conjugate sample-then-retarget chain, not the retarget-then-sample phase order of $K = RA$. Specifically,

$$\frac{\lambda}{m(P)} \xrightarrow{A} \frac{\eta}{m(P)} = \nu(P) \xrightarrow{R} \frac{\lambda}{m(P)}, \qquad \frac{\eta}{m(P)} \xrightarrow{R} \frac{\lambda}{m(P)} \xrightarrow{A} \frac{\eta}{m(P)}. \tag{3.18a}$$

Thus $M(P) = AR$ has stationary law $\lambda/m(P)$, while $K(P) = RA$ has stationary law $\eta/m(P) = \nu(P)$. In one $M(P)$ step, let I be the source drawn by A. The post-sampling set B contains I and has law $\nu(P)$. If the subsequent fair coin in R stops, it chooses the new target W uniformly from that same set B. Consequently

$$\mathbb{P}(W = I \mid B, \text{stop}) = \frac{1}{|B|},$$

and averaging with (3.12) gives

$$\mathbb{P}(\text{stop}, W = I) = \frac{1}{2m(P)}, \qquad \mathbb{P}(W = I \mid \text{stop}) = \frac{1}{m(P)}. \tag{3.18}$$

Thus maximizing fixation is equivalent to minimizing this stationary collision rate.

4 The complete-kernel Hessian

Let $P_\varepsilon = J_n + \varepsilon\delta$ with $\delta \in \mathcal{T}_n$. By linearity,

$$K(P_\varepsilon) = K_0 + \varepsilon\Delta. \tag{4.1}$$

Because every off-diagonal entry of J_n is $1/(n-1)$, each fixed δ admits $\varepsilon_0 > 0$ such that P_ε remains strictly positive off the diagonal for $|\varepsilon| < \varepsilon_0$. Thus all positive-kernel identities from Section 3 apply in the neighborhood used below. All entries are analytic there, and K_0 is irreducible. Put

$$q = H - c_0\mathbf{1}, \qquad G = (I - K_0 + \mathbf{1}\nu_0)^{-1}. \tag{4.2}$$

4.1 Vanishing first variation

Let S average a function on $\mathcal{Y}_n$ over all active states of the same rank $|B|$. Permutation symmetry gives $SK_0 = K_0S$. A direct average of either active move shows

$$S\Delta S = 0. \tag{4.3}$$

Indeed, a uniformly random k-subset of $V \setminus \{v\}$ contains a sample from any row with mean probability k/N; after retargeting the corresponding mean is $(k-1)/N$. More explicitly, if f_k is any rank function, the two branchwise perturbations at (B, v) are

$$\frac{f_{k+1} - f_k}{2} \sum_{i \notin B} \delta_{vi}, \qquad \frac{f_{k-1} - f_k}{2k} \sum_{w \in B} \sum_{i \in B\setminus\{w\}} \delta_{wi}. \tag{4.3a}$$

Under the uniform rank-k orbit their respective row sums are

$$\frac{N-k}{N} \sum_{i \neq v} \delta_{vi} = 0, \qquad \frac{k-1}{N} \sum_{i \neq w} \delta_{wi} = 0. \tag{4.3b}$$

This proves (4.3) without reversibility. Because q and Gq are rank functions and $\nu_0 = \nu_0 S$,

$$\nu_0 \Delta G q = \nu_0 S \Delta S G q = 0. \tag{4.4}$$

Stationary perturbation of the finite chain gives

$$\nu(P_\varepsilon) q = \varepsilon \nu_0 \Delta G q + \varepsilon^2 \nu_0 \Delta G \Delta G q + O(\varepsilon^3). \tag{4.5}$$

Define

$$\mathcal{R}_n^{(2)}(\delta) = \nu_0 \Delta G \Delta G q. \tag{4.6}$$

Equations (3.12), (3.17), and (4.4)–(4.6) already prove expansion (2.6); it remains to prove that $\mathcal{R}_n^{(2)}$ is positive definite on the full tangent space.

4.2 The three irreducible tangent sectors

For $\delta \in \mathcal{T}_n$, let

$$s_j = \sum_i \delta_{ij}, \qquad \sum_j s_j = 0, \tag{4.7}$$

and define the standard embedding

$$E(s)_{ij} = \frac{s_i + (n-1)s_j}{n(n-2)} \quad (i \neq j), \qquad E(s)_{ii} = 0. \tag{4.8}$$

Then $E(s)\mathbf{1} = 0$ and its column-sum vector is s. Consequently $B = \delta - E(s)$ has zero row and column sums, and

$$\delta = E(s) + \frac{B + B^{\mathsf{T}}}{2} + \frac{B - B^{\mathsf{T}}}{2}. \tag{4.9}$$

The three terms are Frobenius-orthogonal. Their dimensions are

$$n-1, \qquad \frac{n(n-3)}{2}, \qquad \frac{(n-1)(n-2)}{2}, \tag{4.10}$$

which sum to $n(n-2) = \dim \mathcal{T}_n$. They are, respectively, the standard, symmetric row/column-balanced, and antisymmetric balanced irreducible S_n-modules. At $n = 3$ the symmetric module has dimension zero.

For completeness, the symmetric off-diagonal matrices form the two-subset permutation module $\mathbf{1} \oplus [n-1,1] \oplus [n-2,2]$; its row-balanced kernel is $[n-2,2]$. The antisymmetric balanced kernel is $\bigwedge^2[n-1,1] = [n-2,1,1]$, while $E(s)$ realizes $[n-1,1]$. Thus the three summands in (4.9) are pairwise nonisomorphic and occur with multiplicity one, including the stated $n = 3$ degeneration.

The quadratic form (4.6) is invariant under simultaneous relabeling of targets and sources. Multiplicity-freeness therefore makes the three spaces orthogonal for $\mathcal{R}_n^{(2)}$ as well, and $\mathcal{R}_n^{(2)}$ acts by one scalar on each. The balanced subspace is exactly

$$\mathcal{B}_n = \mathcal{S}_n \oplus \mathcal{A}_n, \tag{4.10b}$$

where $\mathcal{S}_n$ and $\mathcal{A}_n$ denote the symmetric- and antisymmetric-balanced summands in (4.9). The standard module $E(s)$ is precisely the complementary column-imbalance direction.

Theorem 9 (sector positivity)**.** *The standard scalar is positive for every $n \geq 3$, the symmetric-balanced scalar is positive for every $n \geq 4$, and the antisymmetric-balanced scalar is positive for every $n \geq 3$.*

The proof is in Appendix A. We summarize its architecture. For the standard sector, a signed three-channel quotient separates positive and negative orientations. Schur elimination turns each completed negative excursion into a nonnegative re-entry operator. Exact first-phase barriers and a uniform row-sum contraction control the entire alternating resolvent. For the antisymmetric sector, a one-feature reduction sends a rank Poisson gradient through a positive decreasing recurrence; the second perturbation is then a sum of nonnegative sampling-without-replacement terms, at least one strict. For the symmetric sector, the exact quotient has a good/bad block form

$$\begin{pmatrix} S & C \\ -D & Q \end{pmatrix} \tag{4.11}$$

with S, C, D, Q nonnegative. Schur elimination converts every completed bad excursion into one factor of a nonnegative return operator A. Explicit positive phase barriers dominate the alternating resolvent $(I + A)^{-1}$. The computer-assisted part is finite and exact: rational stationary solves cover $3 \leq N \leq 39$, and rational phase-margin checks cover $40 \leq N \leq 287$. Coefficientwise polynomial and discriminant certificates prove the remaining range $N \geq 288$. Appendix A states the finite computation as a formal lemma and identifies its exact verifier.

For orientation, the first sector eigenvalues are

$$
\begin{array}{c|ccc}
n & \lambda_{\text{std}} & \lambda_{\text{sym}} & \lambda_{\text{anti}} \\
\hline
3 & 1/11 & --- & 1/9 \\
4 & 87/640 & 3/208 & 57/640 \\
5 & 8585/57314 & 359/26660 & 143/2100.
\end{array}
\tag{4.12}
$$

These are the Frobenius-normalized quadratic eigenvalues; they are checks, not a finite extrapolation.

Proof of Theorem 1. Theorem 9 and the decomposition (4.9) imply $\mathcal{R}_n^{(2)}(\delta) > 0$ for every nonzero $\delta \in \mathcal{T}_n$. Equations (3.12) and (4.5) give

$$\frac{1}{n\rho_{\text{dB}}(P_\varepsilon, 2)} = c_0 + \varepsilon^2 \mathcal{R}_n^{(2)}(\delta) + O(\varepsilon^3). \tag{4.13}$$

Differentiation yields (2.8). To obtain a neighborhood rather than only directional statements, restrict the analytic fixation map to a small Frobenius ball around J_n in the affine space $J_n + \mathcal{T}_n$, chosen inside the strictly positive off-diagonal region. The smallest $\mathcal{R}_n^{(2)}$-eigenvalue on the Frobenius unit sphere is positive. Taylor's theorem with a uniform third-derivative bound on a smaller compact ball makes the negative quadratic term dominate the remainder. Hence every other kernel in that ball has strictly smaller fixation probability. □

Remark 10. *The neighborhood in the last paragraph depends on n. No statement here prevents a population sequence from approaching a singular face, or from remaining a nonperturbative distance from J_n as $n \to \infty$.*

5 Fixed structures at strong selection

We now prove Theorem 2, Corollary 3, and Proposition 4.

5.1 Complete-graph baseline

Let ϕ_k be complete-graph fixation from k mutants and let γ_k be the ratio of the probabilities of decreasing and increasing k. First-step equations give

$$\phi_1 = \left(1 + \sum_{j=1}^{n-1} \prod_{k=1}^{j} \gamma_k\right)^{-1}. \tag{5.1}$$

Under dB updating,

$$\gamma_k = \frac{n - 1 + (r-1)k}{r\{n - 1 + (r-1)(k-1)\}}, \tag{5.2}$$

so the product telescopes and yields (2.4). For $n \geq 3$,

$$\rho_{\text{dB}}(J_n, r) = \frac{n-1}{n} - \frac{n-1}{nr} + O(r^{-2}). \tag{5.3}$$

We use the following standard finite-state observation.

Lemma 11 (finite-state perturbation)**.** *Let a finite absorbing transition matrix depend analytically on ε near zero. If the transient restriction at $\varepsilon = 0$ has spectral radius below one, then all absorption probabilities are analytic near zero. More generally, if a limiting reachable transient class has a bounded fundamental matrix and one-step leakage is $O(\varepsilon)$, then leakage before absorption is $O(\varepsilon)$.*

Proof. Write the transient absorption equation as $(I - Q(\varepsilon))h = b$. The inverse exists near zero and is analytic by the adjugate formula. On a restricted limiting class, its bounded fundamental matrix times the leakage vector bounds the probability of leaving before absorption. $\square$

5.2 Complete directed support

Assume $n \geq 3$ and $w_{uv} > 0$ for $u \neq v$. Put

$$\varepsilon = r^{-1}, \qquad a_{uv} = \frac{d_v^- - w_{uv}}{w_{uv}}. \tag{5.4}$$

Let $q_S(\varepsilon)$ be extinction probability from mutant set S. At $\varepsilon = 0$, every set with at least two mutants fixates, while $q_{\{i\}}(0) = 1/n$. The limiting transient chain absorbs, so Lemma 11 gives analyticity.

Differentiate the exact first-step system. If $3 \leq |S| \leq n-1$, a first-order loss still leaves at least two mutants, whose zeroth-order extinction is zero. The derivative therefore satisfies

$$(n - |S|)q_S'(0) = \sum_{v \notin S} q_{S\cup\{v\}}'(0). \tag{5.5}$$

Backward induction from $q_V'(0) = 0$ gives

$$q_S(\varepsilon) = O(\varepsilon^2), \qquad |S| \geq 3. \tag{5.6}$$

For a pair $S = \{i, j\}$, loss of target i has probability

$$\frac{1}{n}\frac{\varepsilon(d_i^- - w_{ji})}{w_{ji} + \varepsilon(d_i^- - w_{ji})} = \frac{\varepsilon a_{ji}}{n} + O(\varepsilon^2), \tag{5.7}$$

and similarly for target j. The leading self-loop mass is $2/n$, the leading gain mass is $(n-2)/n$, and a resulting singleton has extinction $1/n + O(\varepsilon)$. Hence

$$q_{\{i,j\}}(\varepsilon) = \frac{a_{ij} + a_{ji}}{n(n-2)}\varepsilon + O(\varepsilon^2). \tag{5.8}$$

From singleton $\{i\}$, death of resident target v creates the pair $\{i, v\}$ with probability

$$t_{iv} = \frac{1}{n}\frac{w_{iv}}{w_{iv} + \varepsilon(d_v^- - w_{iv})} = \frac{1}{n}(1 - a_{iv}\varepsilon) + O(\varepsilon^2). \tag{5.9}$$

Death of i causes extinction with probability $1/n$. Removing holding probabilities gives

$$\left(\frac{1}{n} + \sum_{v \neq i} t_{iv}\right) q_{\{i\}} = \frac{1}{n} + \sum_{v \neq i} t_{iv} q_{\{i,v\}}. \tag{5.10}$$

Define

$$O_i = \sum_{v \neq i} a_{iv}, \qquad I_i = \sum_{u \neq i} a_{ui}. \tag{5.11}$$

Substitution of (5.8) into (5.10) yields

$$q_{\{i\}}(\varepsilon) = \frac{1}{n} + \frac{\varepsilon}{n^2}\left[O_i + \frac{O_i + I_i}{n-2}\right] + O(\varepsilon^2). \tag{5.12}$$

Since $\sum_i O_i = \sum_i I_i =: T_{\mathrm{dir}}(W)$, uniform averaging gives

$$\rho_{\mathrm{dB}}(W, r) = \frac{n-1}{n} - \frac{T_{\mathrm{dir}}(W)}{n^2(n-2)r} + O(r^{-2}), \tag{5.13}$$

where

$$T_{\mathrm{dir}}(W) = \sum_v \sum_{u \neq v} \frac{d_v^- - w_{uv}}{w_{uv}}. \tag{5.14}$$

For each target column, the elementary identity

$$d_v^- \sum_{u \neq v} \frac{1}{w_{uv}} - (n-1)^2 = \sum_{\substack{u<z \\ u,z \neq v}} \frac{(w_{uv} - w_{zv})^2}{w_{uv} w_{zv}} \tag{5.15}$$

gives

$$T_{\mathrm{dir}}(W) - n(n-1)(n-2) = \mathcal{E}_{\mathrm{dir}}(W). \tag{5.16}$$

Equations (5.3), (5.13), and (5.16) prove Theorem 2. Equality in (5.15) forces each incoming column to be constant. Such column constants cancel from every parent competition, so the full chain is exactly the J_n chain.

5.3 Noncomplete supports and fixed-graph closure

If the positive directed support is strongly connected but noncomplete, Theorem 1 of Tkadlec et al. (2020) proves eventual strict dB suppression. It remains to remove strong connectivity.

Consider the condensation DAG. If it has at least two source strongly connected components, then a uniformly initialized singleton leaves at least one source component entirely resident and unable to receive mutant ancestry; fixation is impossible. Otherwise let $C \subsetneq V$ be the unique source component. Initial mutants outside C cannot fixate. For $i \in C$, let s_i^+ be its positive out-support size. While i is the only mutant, its chance to make a first gain before dying is at most $s_i^+/(s_i^+ + 1)$. Thus

$$\rho_{\mathrm{dB}}(W, r) \leq \frac{1}{n} \sum_{i \in C} \frac{s_i^+}{s_i^+ + 1} \leq \frac{(n-1)^2}{n^2}. \tag{5.17}$$

The complete baseline tends to $(n-1)/n$, leaving limiting gap $(n-1)/n^2$. Combining this case, the cited strongly connected noncomplete-support theorem, and Theorem 2 proves Corollary 3.

5.4 Undirected incomplete support

Fix an initial mutant i in a connected undirected support. At $r = \infty$, death of i causes extinction, while death of any one of its s_i neighbors creates an adjacent mutant pair. After deleting holding steps, the probability of reaching a pair first is $s_i/(s_i + 1)$. Once an adjacent pair exists, the mutant set remains support-connected; every mutant has a mutant neighbor and cannot be replaced by a resident in the limiting chain, while every boundary resident death produces monotone growth. The limiting chain therefore reaches fixation almost surely from the pair. Lemma 11 controls $O(1/r)$ leakage from this limiting class. Uniform averaging proves (2.11); elementary subtraction proves (2.12).

6 Global all-fitness slices at small order

Local curvature at $r = 2$ does not imply global maximality. At order three, however, the full all-fitness comparison admits an exact centered sum-of-squares certificate.

Let an undirected triangle have positive edge weights (a, b, c). If F_i is fixation when i is the unique mutant and G_i when i is the unique resident, deleting holding probabilities gives six linear equations. For distinct i, j, k, put

$$u_{ij} = \frac{rw_{ij}}{rw_{ij} + w_{jk}}, \qquad v_{ij} = \frac{w_{ij}}{rw_{jk} + w_{ij}}. \tag{6.1}$$

Then

$$(1 + u_{ij} + u_{ik})F_i = u_{ij}G_k + u_{ik}G_j, \tag{6.2}$$

$$(1 + v_{ij} + v_{ik})G_i = 1 + v_{ij}F_k + v_{ik}F_j. \tag{6.3}$$

The coefficient matrix is a strictly diagonally dominant M-matrix, so its determinant is positive.

Put $s_1 = a + b + c$, $s_2 = ab + ac + bc$, and $s_3 = abc$, and define

$$\begin{aligned} A &= 3s_3(s_1s_2 - 9s_3), \\ D &= 12s_1^3s_3 - 45s_1s_2s_3 + 4s_2^3 - 27s_3^2, \\ E &= 4s_2(3s_1^2s_2 - 3s_1s_3 - 8s_2^2), \\ \mathcal{H}_r &= A(r-1)^4 + Dr(r-1)^2 + Er^2. \end{aligned} \tag{6.4}$$

Exact elimination of (6.2)–(6.3) gives

$$\rho_{\mathrm{dB}}(W, r) - \rho_{\mathrm{dB}}(J_3, r) = -\frac{r(r-1)\mathcal{H}_r}{3(r+1)\mathcal{P}_r}, \tag{6.5}$$

where $\mathcal{P}_r > 0$ is the cleared determinant. More explicitly, if M is the coefficient matrix of (6.2)–(6.3) and

$$L = \prod_{\substack{p,q \in \{a,b,c\} \\ p \neq q}} (rp + q) > 0,$$

then $\mathcal{P}_r = L \det(M)/3 > 0$.

To sign the numerator, set

$$U = s_1^2 - 3s_2, \quad V = s_2^2 - 3s_1s_3, \quad W = s_1s_2 - 9s_3, \quad Z = s_2^3 - 27s_3^2. \tag{6.6}$$

Writing $X = ab$, $Y = ac$, $Z_0 = bc$ gives

$$\begin{aligned} 2U &= (a-b)^2 + (a-c)^2 + (b-c)^2, \\ 2V &= (X-Y)^2 + (X-Z_0)^2 + (Y-Z_0)^2, \\ W &= c(a-b)^2 + b(a-c)^2 + a(b-c)^2, \\ Z &= s_2V + 3\{Z_0(X-Y)^2 + Y(X-Z_0)^2 + X(Y-Z_0)^2\}. \end{aligned} \tag{6.7}$$

The coefficient identities

$$A = 3s_3W, \qquad D = 12s_1s_3U + 3s_2V + Z, \qquad E = 4s_2(3s_2U + V) \tag{6.8}$$

show that $A, D, E \geq 0$. If the weights are nonuniform then $U > 0$, hence $E > 0$ and $\mathcal{H}_r > 0$ for $r > 1$. Equation (6.5) proves Theorem 5, including its equality class.

Appendix B gives analogous exact certificates for the $1+3$ and $2+2$ invariant positive weighted K_4 families. Those families are included because they test the first genuinely higher-dimensional subset chains while retaining exact orbit reductions. We make no claim that they classify all weighted four-vertex graphs.

7 Implications, limitations, and reproducibility

7.1 A necessary condition for asymptotic universal amplification

The fixed-graph result does not interchange n and r. We first record the monotonicity needed for a useful support condition on graph sequences.

Lemma 12 (fitness monotonicity)**.** *For every loopless row-stochastic replacement kernel P and every initial mutant set, the dB fixation probability is nondecreasing in mutant fitness $r > 0$.*

Proof. For $0 \le x \le 1$,

$$\frac{\partial f_r(x)}{\partial r} = \frac{x(1-x)}{[1+(r-1)x]^2} \ge 0, \qquad \frac{\partial f_r(x)}{\partial x} = \frac{r}{[1+(r-1)x]^2} > 0.$$

Couple chains with $r \le r'$ by using the same dead target v and the same uniform threshold U at every update, declaring the replacement mutant when $U \le f_r(P_{vS})$. If their current mutant sets satisfy $S \subseteq T$, then $P_{vS} \le P_{vT}$ and the two displayed monotonicities preserve the inclusion after the update. Starting both chains from the same set, fixation of the r-chain therefore implies fixation of the r'-chain. □

Lemma 12 and Proposition 4 give, for every finite connected undirected graph,

$$\rho_{\mathrm{dB}}(W, r) \le 1 - \frac{1}{n} \sum_i \frac{1}{s_i + 1}. \tag{7.1}$$

Proposition 13 (support degree must diverge)**.** *Let W_n be connected undirected weighted graphs of order $n \to \infty$. If, for every fixed $r > 1$, W_n is eventually a dB amplifier, then*

$$\frac{1}{n} \sum_i \frac{1}{s_i + 1} \longrightarrow 0. \tag{7.2}$$

Consequently the support degree of a uniformly chosen vertex tends to infinity in probability.

Proof. At fixed $R > 1$, eventual amplification and (7.1) give

$$\limsup_{n\to\infty} \frac{1}{n} \sum_i \frac{1}{s_i + 1} \le \lim_{n\to\infty} \{1 - \rho_{\mathrm{dB}}(J_n, R)\} = \frac{1}{R}.$$

Let $R \to \infty$. For fixed K, the fraction with $s_i \le K$ is at most $(K+1)$ times the average in (7.2). □

This excludes uniformly bounded-support families, but not supports completed by arbitrarily weak positive edges, diffuse perturbations, mesoscopic modules, or singular population-dependent kernels.

7.2 What the local theorem does and does not say

Theorem 1 proves strict negative fixation curvature in every nonzero normalized directed direction at fixed order. In particular, neither column imbalance nor directed circulation escapes the local maximum. The theorem does not supply a radius uniform in n, global concavity along rays, or a path from an arbitrary kernel to J_n on which fixation is monotone. The global conjecture

$$\rho_{\mathrm{dB}}(P, 2) \le \rho_{\mathrm{dB}}(J_n, 2) \tag{7.3}$$

for every kernel P induced by an undirected weighting remains open.

7.3 Exact verification and proof status

The archive contains independent exact implementations for the two theorem axes. The strong-selection suite constructs literal subset-state chains, checks row normalization and complete-graph lumpability, differentiates the absorption equations, verifies the incoming-column sum of squares, and replays the triangle and K_4 certificates. The fitness-two suite independently builds the marked and active chains, verifies the collision normalization, reconstructs the rank Poisson equation, checks the tangent decomposition, and replays all three irreducible-sector certificates.

The standard sector has an analytic tail certificate for $N \geq 10$ and exact rational closure for $2 \leq N \leq 9$. The symmetric sector uses exact solves for $3 \leq N \leq 39$, exact rational phase margins for $40 \leq N \leq 287$, and a coefficient/discriminant certificate for $N \geq 288$. All arithmetic in these finite ranges is rational. The verifiers discharge the finite computer-assisted portions; sampled numerical values are never used to extend a quantifier.

The one-command replay is

```
./replay.sh
```

from the paper package. The package-level integration audit separately checks the complete active stationary law, the full tangent decomposition, all three physical sector normalizations, the strong-selection column identity, and conversion from inverse-mean to fixation curvature.

Statements and Declarations

Data and code availability. No empirical data are used. Complete LaTeX source, exact rational and symbolic verifiers, coefficient certificates, build scripts, and adversarial audit reports accompany this manuscript in the development repository https://github.com/AlecKriebel/Math. The earlier strong-selection and low-order version is archived at https://doi.org/10.5281/zenodo.21753405; that record does not contain the new fitness-two local theorem. The accompanying standalone source bundle freezes every in-repository dependency needed to replay this manuscript. Its release command prints the archive checksum, while the accompanying reproduction note records the clean-room test. No persistent identifier has yet been assigned to this consolidated version. The final availability statement must name its versioned deposit once the human-authorized preprint release exists.

Funding. No external funding supported this work.

Competing interests. The author declares no competing interests.

Author contributions. A.K. determined the research questions, scope, and claim boundaries, reviewed the manuscript and reproducibility artifacts, and accepts responsibility for the work.

Ethics approval and consent. Not applicable. This study uses no human participants, animals, or empirical biological data.

AI-assisted research disclosure. Generative-AI systems were used substantively in mathematical exploration, derivation, adversarial proof analysis, software development, exact verification, literature support, visualization, and manuscript preparation. The named human author determined the scope and claims, reviewed the final manuscript and public artifacts, and accepts responsibility for the work. All exact certificates are supplied for independent replay, and numerical evidence is explicitly separated from proof. AI systems are not listed as authors.

A Hessian-sector certificates for every population size

This appendix proves Theorem 9. Throughout, $N = n - 1$ and

$$\pi_k = \frac{\binom{N-1}{k-1}}{2^{N-1}}, \qquad 1 \le k \le N, \tag{A.1}$$

is the complete active-rank law. Its rank chain has

$$R_{k,k+1} = \frac{N-k}{2N}, \qquad R_{k,k-1} = \frac{k-1}{2N}. \tag{A.2}$$

Let h solve $(I - R)h = k^{-1} - c_0$ modulo constants and put $d_k = h_k - h_{k+1}$ for $1 \le k < N$, with $d_0 = d_N = 0$. Subtracting adjacent Poisson equations gives

$$(N-k)d_k - (k-1)d_{k-1} = 2N\left(\frac{1}{k} - c_0\right), \qquad 1 \le k < N. \tag{A.3}$$

Solving this recurrence gives, for $1 \le k < N$,

$$\frac{2}{k} - d_k = \frac{2^{2-N}\sum_{r=k}^{N-1}\binom{N-1}{r}}{(N-1)\binom{N-2}{k-1}}, \qquad \frac{2(N-2)}{Nk} \le d_k \le \frac{2}{k}. \tag{A.3a}$$

For the lower bound, after clearing the positive denominator it is enough to show

$$Nk\sum_{r=k}^{N-1}\binom{N-1}{r} \le 2^N(N-1)\binom{N-2}{k-1}. \tag{A.3b}$$

For $k \le N/2$, use the full binomial sum and $\binom{N-2}{k-1} \ge N-2$ away from the immediate endpoint $k = 1$; for $k \ge 2, N \ge 4$, the remaining comparison follows from $Nk \le N^2/2 \le 2(N-1)(N-2)$. The case $k = 1$ is immediate. For $k > N/2$, symmetry and monotonicity of the lower half of the binomial row give

$$\sum_{r=k}^{N-1}\binom{N-1}{r} \le (N-k)\frac{N-1}{k}\binom{N-2}{k-1},$$

and $N(N-k) \le 2^N$ completes the proof. The order $N = 2$ is direct.

A.1 Standard column-imbalance directions

Let $\xi \in \mathbb{R}^{N+1}$ have coordinate sum zero and let $E(\xi)$ be the standard embedding in (4.8). By S_{N+1}-invariance and irreducibility of the standard module, it is enough to calculate the scalar for the canonical vector $\xi = e_x - e_y$; the resulting ratio then holds for every zero-sum ξ. Distinguish the labels x, y. The signed two-label quotient has positive channels

$$\begin{array}{r|l} P_k & v = x,\ x, y \notin B, \quad 1 \le k < N, \\ Q_k & v = x,\ y \in B, \quad 1 \le k \le N, \\ R_k & v \notin \{x, y\},\ x \in B,\ y \notin B, \quad 1 \le k < N, \end{array}$$

with the negative orientation obtained by exchanging x, y. Use the channel orders

$$\mathcal{G} = (P_1, \ldots, P_{N-1}, R_1, \ldots, R_{N-1}), \qquad \mathcal{Q} = (Q_1, \ldots, Q_N).$$

In these coordinates the operator is

$$H = \begin{pmatrix} S & C \\ -D & Q \end{pmatrix}, \tag{A.4s}$$

where all four displayed blocks are nonnegative. The nonzero signed rows are

$$P_k \mapsto \frac{k}{2N} P_k + \frac{N-k-1}{2N} P_{k+1} + \frac{1}{2N} Q_{k+1}, \tag{A.5s}$$

$$\begin{aligned} Q_k \mapsto \frac{k^2-1}{2kN} Q_k + \frac{N-k}{2N} Q_{k+1} - \frac{k-1}{2kN} P_{k-1} - \frac{N-k}{2kN} P_k \\ - \frac{(k-1)^2}{2kN} R_{k-1} - \frac{(k-1)(N-k)}{2kN} R_k, \end{aligned} \tag{A.6s}$$

and

$$\begin{aligned} R_k \mapsto \left\{ \frac{k}{2N} + \frac{(k-1)(N-k)}{2kN} \right\} R_k + \frac{N-k-1}{2N} R_{k+1} + \frac{(k-1)^2}{2kN} R_{k-1} \\ + \frac{1}{2kN} Q_k + \frac{k-1}{2kN} P_{k-1} + \frac{N-k}{2kN} P_k. \end{aligned} \tag{A.7s}$$

Terms outside their physical ranges are absent. Define the binomial row

$$\sigma(R_k) = \frac{\binom{N-2}{k-1}}{2^{N-2}}, \qquad \sigma(P_k) = 0, \qquad \sigma(Q_k) = 0, \tag{A.8s}$$

and the physical reward

$$\begin{aligned} \gamma(P_k) &= k d_k, \\ \gamma(R_k) &= N d_k + \frac{(N+1)(k-1)}{k} d_{k-1}, \\ \gamma(Q_k) &= -q_k, \quad q_k = (N-k) d_k + \frac{(N+1)(k-1)}{k} d_{k-1} > 0. \end{aligned} \tag{A.9s}$$

The standard-sector scalar is

$$\Phi_N = \sigma (I-H)^{-1} \gamma. \tag{A.10s}$$

This is the physical, rather than an auxiliary, reward. Indeed the first perturbation has feature coefficients

$$a_k = \frac{k d_k}{2(N+1)(N-1)}, \qquad b_k = \frac{N d_k}{2(N+1)(N-1)} + \frac{(k-1) d_{k-1}}{2k(N-1)}, \tag{A.11s}$$

whose P, R, Q values are $a_k, b_k, a_k - b_k$. The averaged second perturbation annihilates the a feature and has b_k coefficient $\sigma(R_k)/[2(N+1)]$. Therefore

$$\boxed{\frac{\mathcal{R}_n^{(2)}(E(\xi))}{\|\xi\|^2} = \frac{\Phi_N}{4(N+1)^2(N-1)}.} \tag{A.12s}$$

The table in Section 4 instead uses Frobenius normalization on the embedded matrix. Direct summation of (4.8) gives

$$\|E(\xi)\|_F^2 = \frac{n-1}{n(n-2)} \|\xi\|^2, \tag{A.12n}$$

and therefore the Frobenius-normalized standard eigenvalue is $\Phi_N/[4N(N+1)]$. For example, the canonical $n=3$ ratio $\mathcal{R}_3^{(2)}(E(\xi))/\|\xi\|^2 = 2/33$ becomes the table value $1/11$.

It remains to prove $\Phi_N > 0$. Put

$$R_S = (I - S)^{-1}, \quad R_Q = (I - Q)^{-1}, \quad A = R_S C R_Q D,$$

where the displayed rows give $S\mathbf{1} \le (N-1)\mathbf{1}/N$ and $(Q\mathbf{1})_k = 1/2 - 1/(2kN)$. Thus both inverses are convergent Neumann series and are entrywise nonnegative. Also put

$$h = R_Q q, \qquad r_0 = \gamma_S - Ch, \qquad f_0 = R_S r_0. \tag{A.13s}$$

Schur elimination of the bad channel gives the exact alternating-phase identity

$$\Phi_N = \sigma(I + A)^{-1} f_0. \tag{A.14s}$$

For $N \ge 6$, define

$$W_1 = 2N^2 d_1, \quad W_2 = 2N d_1, \quad W_k = \frac{4N(k-1)}{k} d_{k-1} \quad (3 \le k \le N). \tag{A.15s}$$

Then $(I - Q)W \ge q$. At ranks one and two the cleared residuals are $2(N^2 - N + 1)d_1$ and

$$\frac{3N-4}{2} d_1 - \frac{7(N-2)}{3} d_2 \ge \frac{2(N-2)(N-6)}{3N}.$$

For $k \ge 3$ the residual is $A_k d_{k-1} - B_k d_k$, with

$$A_k = \frac{(k-1)(3kN - 2k^2 - k + 2)}{k^2}, \qquad B_k = \frac{(3k+1)(N-k)}{k+1}.$$

Using (A.3a), it is at least $2P(N,k)/[Nk^2(k+1)]$. On setting $a = k-3$ and $m = N-k$,

$$\begin{aligned} P &= a^4 + a^3 m + 10a^3 + 5a^2 m + 37a^2 + 60a \\ &\quad + 2am(m-1) + 6m^2 - 22m + 32 > 0. \end{aligned} \tag{A.16s}$$

the last quadratic has discriminant -284. Thus $h \le W$. Direct substitution now gives

$$\begin{array}{c|c} P_1 & 0 \\ P_k & k(k-1)d_k/(k+1), \quad 2 \le k < N \\ R_1 & 0 \\ R_2 & Nd_2 + Nd_1/2 \\ R_k & Nd_k + \frac{k-1}{k}(N + 1 - 2/k)d_{k-1}, \quad 3 \le k < N \end{array} \tag{A.17s}$$

for the coordinates of $\gamma_S - CW$. Hence $f_0 \ge 0$.

The bad exit and retention row sums satisfy

$$D\mathbf{1} = \frac{N-1}{2N}\mathbf{1}, \qquad (Q\mathbf{1})_k = \frac{1}{2} - \frac{1}{2kN}. \tag{A.18s}$$

Consequently $R_Q D\mathbf{1} \le \mathbf{1}$. With

$$z(P_k) = 1/N, \qquad z(R_k) = 2/(N+k),$$

one has $(I - S)z \ge C\mathbf{1}$. The P_k gap is $1/(2N^2)$; after $a = k-1, m = N-k$, the cleared R_k gap is

$$\begin{aligned} Z &= 4a^4 + 14a^3 m + 18a^3 + 14a^2 m^2 + 40a^2 m + 30a^2 \\ &\quad + 4am^3 + 24am^2 + 36am + 20a + 3m^3 + 8m^2 + 9m + 4 > 0. \end{aligned} \tag{A.19s}$$

Applying the positive inverses proves

$$A\mathbf{1} \le \frac{2}{N+1}\mathbf{1}. \tag{A.20s}$$

Two elementary barriers control the first phase. The supersolution $u(P_k) = 4, u(R_k) = 8N$ gives $0 \le f_0 \le 8N\mathbf{1}$. Indeed its P residual is $2(N+1)/N \ge kd_k$, while its R residual exceeds the upper bound $2(2N+1)/k$ for $\gamma(R_k)$ by $2(4Nk-4N+1)/(Nk)$. Separately, $q_k \le 2N$ and (A.18s) imply $h \le 4N\mathbf{1}$. For $N \ge 7$, the subsolution $\ell(P_k) = 0, \ell(R_k) = 2N$ then gives

$$f_0(R_k) \ge 2N, \qquad \sigma f_0 \ge 2N. \tag{A.21s}$$

At $k = 1$ the margin is $2N - 6 - (N+1) \ge 0$; at $k \ge 2$ it is $[3N - 7 - 2k - 4/N]/k \ge 0$.

Set $c_N = 2/(N+1)$. Equations (A.20s)–(A.21s) and the Neumann series for $(I+A)^{-1}$ yield, for $N \ge 10$,

$$\Phi_N \ge 2N - 8N \sum_{j\ge1} c_N^j = \frac{2N(N-9)}{N-1} > 0. \tag{A.22s}$$

The remaining exact values are

N	Φ_N	N	Φ_N
2	24/11	6	5758562957/248448224
3	261/40	7	141339691089527/4988552903680
4	343400/28657	8	15468663676289/466560376100
5	2268275/128288	9	19782952499295763/524622207176704.

(A.23s)

This proves the standard-sector sign for every population size. The supplementary verifier independently reconstructs the quotient, normalization, barriers, shifted polynomials, and every value in (A.23s).

A.2 Antisymmetric balanced directions

Let $\delta = -\delta^{\mathsf{T}}$ have zero row sums and set

$$x(B, v) = \sum_{i\in B} \delta_{vi}. \tag{A.4}$$

Direct expansion of the active moves gives

$$\Delta h(B, v) = \frac{d_k}{2} x(B, v), \qquad k = |B|. \tag{A.5}$$

Lemma 14 (strict rank-Poisson gradients)**.** *The Poisson gradients satisfy*

$$d_1 > d_2 > \cdots > d_{N-1} > 0. \tag{A.6}$$

Proof. Realize the chain (A.2) on $N-1$ Boolean coordinates. At each step choose uniformly among these coordinates and one dummy slot; a genuine coordinate, when chosen, is refreshed to an independent fair bit, whereas the dummy leaves the configuration unchanged. If K_t is one plus the number of occupied coordinates, its transition probabilities are exactly (A.2). For $f(k) = k^{-1} - c_0$, stationarity gives $\pi f = 0$, and finite ergodicity gives the convergent Poisson representation, modulo an irrelevant constant,

$$h_k = \sum_{t\ge0} \mathbb{E}_k f(K_t).$$

Couple configurations of ranks k and $k+1$ that differ only at a distinguished bit a, using the same chosen slots and fair refreshes. Until the first refresh of a, write X_t for the number of occupied common coordinates. The forcing difference is

$$g(X_t) := \frac{1}{X_t+1} - \frac{1}{X_t+2} > 0;$$

after that refresh the two chains coalesce. Termwise summation of the Poisson series therefore gives $d_k > 0$.

For $1 \le k \le N-2$, choose another bit b and a set S of $k-1$ occupied bits, disjoint from a, b. Start four chains from

$$S, \qquad S \cup \{a\}, \qquad S \cup \{b\}, \qquad S \cup \{a, b\},$$

and again use identical refreshes. The first pair represents d_k and the second represents d_{k+1}. The common configuration of the second pair contains that of the first until b is refreshed, after which the corresponding common configurations coincide. Since g is strictly decreasing, every summand for d_k dominates the corresponding summand for d_{k+1}, and the inequality is strict already at time zero. Hence $d_k > d_{k+1}$, proving the lemma. □

For any rank sequence f_k,

$$K_0\{f_k x(B,v)\} = \frac{k f_k + (N-k-1) f_{k+1}}{2N} x(B,v). \tag{A.7}$$

Thus $G\Delta h = r_k x$, where $r_N = 0$ and

$$(2N-k) r_k - (N-k-1) r_{k+1} = N d_k. \tag{A.8}$$

Backward induction using (A.6) gives

$$0 \le r_{k+1} < r_k \le \frac{N}{N+1} d_k. \tag{A.9}$$

Put $T = \|\delta\|_F^2$. Sampling without replacement and applying Δ a second time yields

$$\begin{aligned}\mathbb{E}_k[\Delta(r_k x)] = \frac{T}{2nN}\Big[&\frac{k(N-k)}{N-1}(r_k - r_{k+1}) + (N-k) r_{k+1} \\ &+ \frac{(k-1)(N-k+1)}{N-1}(r_{k-1} - r_k) + (N-k+1) r_k\Big].\end{aligned} \tag{A.10}$$

The coefficient of the absent $r_0 - r_1$ term is zero. Every displayed term is nonnegative by (A.9), and the total is strict when $T > 0$. Averaging with (A.1) proves positivity of the antisymmetric sector for every $N \ge 2$.

A.3 Symmetric balanced directions

For a symmetric balanced δ, put

$$x(B,v) = \sum_{i \in B} \delta_{vi}, \qquad z(B) = \sum_{\substack{w,i \in B \\ w \neq i}} \delta_{wi}. \tag{A.11}$$

Functions in this sector have the form $a_k x + b_k z$. Transpose the exact two-channel rank operator and split its positive and negative channels:

$$H = K^{\mathsf{T}} = \begin{pmatrix} S & C \\ -D & Q \end{pmatrix}. \tag{A.12}$$

Every displayed block is entrywise nonnegative. The nonzero entries are

$$\begin{aligned} S_{k,k} &= \frac{k}{2N}, & S_{k,k-1} &= \frac{N-k}{2N}, \\ Q_{k,k} &= \frac{N(k-2)+k}{2kN}, & Q_{k,k-1} &= \frac{N-k-1}{2N}, \\ Q_{k,k+1} &= \frac{k(k-1)}{2(k+1)N}, & D_{k,k-1} &= \frac{1}{N}, \end{aligned} \tag{A.13}$$

and

$$C_{k-1,k} = \frac{k-1}{2kN}, \qquad C_{k,k} = \frac{N-k}{2kN}. \tag{A.14}$$

The a channel is indexed by $1 \le k < N$ and the b channel by $2 \le k < N$; entries outside these ranges are zero. The positive source is

$$s_k^a = \frac{d_k}{2}, \qquad s_k^b = \frac{d_{k-1}}{2k}, \tag{A.15}$$

and the good/bad reward is

$$g_k^a = \frac{\binom{N-2}{k-1}}{2^{N-1}(N+1)}, \qquad g_k^b = -\frac{\binom{N-3}{k-2}}{2^{N-2}(N+1)}. \tag{A.16}$$

The normalization connecting this quotient to the physical Hessian is exact, not merely sign preserving. If $(a,b) = (I-K)^{-1}s$, set $b_1 = a_N = b_N = 0$ and

$$r_k = a_k - \frac{2(k-1)}{N-2} b_k,$$

then the fixed-count orbit identities give

$$\frac{\mathcal{R}_n^{(2)}(\delta)}{\|\delta\|_F^2} = \sum_{k=1}^{N-1} \pi_k \frac{N-k}{(N-1)(N+1)} r_k = g^{\mathsf{T}}(I-K)^{-1}s =: \mathcal{S}_N. \tag{A.17}$$

Here is the labelled normalization valid for every n. Let $c = 1/[nN2^{N-1}]$ and write $\mu = \nu_0\Delta$. For an output state (B,v) of rank k, the four predecessor families contribute, respectively,

$$\frac{c}{2} k x(B,v), \quad \frac{c}{2}(k-1)x(B,v), \quad \frac{c}{2}(N-k)x(B,v), \quad \frac{c}{2}(N-k+1)x(B,v).$$

Thus their sum is the exact labelled current $\mu(B,v) = x(B,v)/(n2^{N-1})$, including the boundary ranks. Define $f(B,v) = a_{|B|}x(B,v) + b_{|B|}z(B)$. Conditional on a uniform rank-k orbit, sampling without replacement gives, with $R_v = \sum_i \delta_{vi}^2$,

$$\mathbb{E}[x^2 \mid v] = \frac{k(N-k)}{N(N-1)} R_v, \qquad \mathbb{E}[xz \mid v] = -\frac{2k(k-1)(N-k)}{N(N-1)(N-2)} R_v. \tag{A.17a}$$

For the cross term, separate the two coincidences in the ordered triple from the all-distinct case; the row and column sums reduce the resulting full sums to $-2R_v$. Summing R_v over v gives $\|\delta\|_F^2$. Therefore

$$\frac{\nu_0 \Delta f}{\|\delta\|_F^2} = \sum_{k=1}^{N-1} \pi_k \frac{N-k}{(N-1)(N+1)} \left(a_k - \frac{2(k-1)}{N-2} b_k \right), \tag{A.17b}$$

which is the first expression in (A.17). The identities

$$\pi_k \frac{N-k}{N-1} = \frac{\binom{N-2}{k-1}}{2^{N-1}}, \qquad \frac{k-1}{N-2}\binom{N-2}{k-1} = \binom{N-3}{k-2}$$

show that the coefficients in (A.17b) are exactly g_k^a, g_k^b. Hence the orbit average is $g^\mathsf{T}(a,b) = g^\mathsf{T}(I-K)^{-1}s$. This is a labelled derivation; the exact replay independently reconstructs it for $4 \le n \le 12$ before taking a sign.

Put $R_S = (I-S)^{-1}$ and $R_Q = (I-Q)^{-1}$. Here S has row sums at most $1/2$, while the displayed tridiagonal Q is substochastic and every communicating class reaches a strict boundary row. Hence both inverses are convergent entrywise-nonnegative Neumann series. Define

$$\begin{aligned} W &= R_Q(-g^b), \quad & r &= g^a - CW, \quad & f_0 &= R_S r, \\ A &= R_S C R_Q D, \quad & \ell &= (s^a)^\mathsf{T} - (s^b)^\mathsf{T} R_Q D, \quad & \mathcal{D} &= (s^b)^\mathsf{T} W. \end{aligned} \tag{A.18}$$

Solving the bad equation first gives the exact Schur identity

$$\mathcal{S}_N = \ell(I+A)^{-1} f_0 - \mathcal{D}. \tag{A.19}$$

One factor of the nonnegative matrix A is one completed visit to the bad channel. The inverse in (A.19) therefore alternates successive re-entries, so a first-excursion estimate alone would not suffice.

We now give the full phase certificate. Put

$$v = R_S g^a, \qquad v_k = t_k g_k^a, \qquad t_1 = \frac{2N}{2N-1}, \quad t_k = \frac{2N + (k-1)t_{k-1}}{2N-k}. \tag{A.20}$$

Induction gives

$$t_k \le \frac{N^2 - k}{(N-2)(N-k)}. \tag{A.21}$$

The base difference is $(5N-1)/[(N-2)(2N-1)]$; after one step and the substitution $m = N-k$, the remaining numerator is $N^3 - N^2 + 2Nm^2 + 2Nm + 2m^3 + 3m^2 + m > 0$.

For $N \ge 24$, let $\overline{W} = (7N/25)(-g^b)$. Direct substitution shows

$$(I-Q)\overline{W} \ge -g^b, \qquad C\overline{W} \le \frac{14}{25} g^a. \tag{A.22}$$

For the first inequality, after clearing positive binomial factors the interior residual is

$$P_N(k) = 21N^2 k + 14N^2 - 64Nk^2 - 57Nk + 50k^3 + 36k^2 - 14k. \tag{A.23}$$

For $N \ge 25$, $\operatorname{disc}_k P_N = -8B_N$, where the coefficients of B_{25+M} are

$$(5733, 736211, 37934833, 982793213, 12893444604, 70866894144, 41021531998). \tag{A.24}$$

Thus P_N has one real, negative root and is positive for $k \ge 0$; at $N = 24$ its exact minimum over the physical ranks is 24. Applying R_Q in (A.22) yields

$$\frac{11}{25} v \le f_0 \le v. \tag{A.25}$$

For $2 \le k < N$, define

$$\widehat{h}_k = \frac{k}{N-2} g^a_{k-1}. \tag{A.26}$$

Equations (A.13) and (A.21) give $(I - Q)\widehat{h} \ge Dv$ and

$$C\widehat{h} \le c_N g^a, \qquad c_N = \frac{2N-5}{2N(N-2)}.$$

Indeed, the interior ratio in the first comparison is $(N^2 - k + 1)/[N(N-2)(N-k+1)]$, and the largest row ratio in the second is at rank $N-2$. At $k = 2$ the first boundary residual is $(N+1)/[N(N-2)]$, equal to the interior formula; at $k = N-1$ it is $(N^2+1)/[2N(N-2)]$, which exceeds the interior value $(N^2 - N + 2)/[2N(N-2)]$. Thus (A.21) dominates every boundary row as well. Consequently

$$Av \le c_N v. \tag{A.27}$$

For the left occupation debt, put

$$Y_k = \frac{2(k+1)}{3k(k-1)} \qquad (2 \le k < N). \tag{A.28}$$

With $B = Q^{\mathsf{T}}$, an interior substitution gives

$$[(I-B)Y]_k - \frac{1}{k(k-1)} = \frac{4Nk + 2N + 2k^3 - k^2 - 5k}{3Nk^2(k-1)(k+1)} > 0, \tag{A.29}$$

while the boundary residuals at $k = 2$ and $k = N-1$ are, respectively, $(5N+7)/(18N) > 0$ and $(N^2 - N + 2)/[3N(N-2)(N-1)^2] > 0$. Since $s^b_k \le 1/[k(k-1)]$, this proves $(s^b)^{\mathsf{T}} R_Q \le Y^{\mathsf{T}}$ and hence $\ell \ge \overline{\ell} > 0$, where

$$\overline{\ell}_j = \frac{3Nj + 3N - 8j - 10}{3Nj(j+1)}. \tag{A.30}$$

The absent bad state above the top rank only increases the final margin.

The bad debt and first phase now satisfy

$$\mathcal{D} \le \sum_{j=1}^{N-2} \frac{4(j+2)}{3(j+1)(N-2)} g^a_j, \qquad \ell f_0 \ge \frac{11}{25} \sum_{j=1}^{N-1} \overline{\ell}_j t_j g^a_j. \tag{A.31}$$

Define

$$\begin{aligned} \beta_N &= \max_{1 \le j \le N-2} \frac{4(j+2)}{3(j+1)(N-2)} \left(\frac{11}{25} \overline{\ell}_j t_j \right)^{-1}, \\ \varepsilon_N &= \frac{25}{11} \frac{c_N}{1 - c_N}. \end{aligned} \tag{A.32}$$

Lemma 15 (finite exact symmetric-sector certificate)**.** *The scalar $\mathcal{S}_N$ is positive for $3 \le N \le 39$. For each of the 248 integer orders $40 \le N \le 287$, $\beta_N + \varepsilon_N < 1$; the smallest certified margin occurs at $N = 40$ and is*

$$1 - \beta_{40} - \varepsilon_{40} = \frac{639304267467075678841}{115369588296792467144716} > 0. \tag{A.33}$$

Computer-assisted proof. The included exact-arithmetic program

`verify_true_inverse_rank_symmetric_phase.py`

directly solves the rational systems for $3 \le N \le 39$, recursively constructs the rational t_j, β_N, and ε_N for all 248 remaining finite orders, asserts every sign and the minimum in (A.33), and uses no floating-point arithmetic. Its SHA-256 is

`b4d45a83ce5f21a1fd3e09403b376e071330290a01affff64711574b69e024bc`.

Successful execution ends with

```
PASS: stationary symmetric inverse-rank phase certificate
```

and prints the three ranges $3{:}39$, $40{:}287$, and $N \ge 288$ separately. The deterministic source bundle contains this exact file and its dependencies. □

For $N \ge 288$, induction in (A.20) gives $t_j \ge N/(N - j + \sqrt{N/2})$. Indeed, substituting the inductive bound leaves the numerator $2u(N - j + u) + u - N$, where $u = \sqrt{N/2}$; it is minimized at $j = N - 1$ and equals $3u > 0$. Since $u \le N/24$ in this range, $t_j \ge 24N/(25N - 24j)$. The bound $\beta_N < 19/20$ follows, after clearing positive factors, from $G_N(j) > 0$, where

$$\begin{aligned} G_N(j) = 1881N^2j + 1881N^2 - 6250Nj^2 - 21278Nj - 10032N \\ + 6000j^3 + 12000j^2 + 10032j + 12540. \end{aligned} \tag{A.34}$$

Its discriminant is $-500D_N$, and the coefficients of D_{288+M} are

$$\begin{gathered} 43034652243, 71940715263252, 50075984929826764, \\ 18577025353519361184, 3873653773133773223808, \\ 430447522448513182675968, 19913301794000751100222464. \end{gathered} \tag{A.35}$$

Thus G_N has one negative real root and is positive on every physical rank. Also $\varepsilon_N < 1/20$ for $N \ge 46$: after clearing its positive denominator this is $22N^2 - 1066N + 2555 > 0$, which holds at $N = 46$ and is increasing thereafter. Therefore $\beta_N + \varepsilon_N < 1$ for all $N \ge 288$.

Finally, (A.25), (A.27), and positivity of ℓ give

$$\left|\ell(I + A)^{-1} f_0 - \ell f_0\right| \le \varepsilon_N \ell f_0.$$

Together with (A.31) this yields

$$\mathcal{S}_N \ge (1 - \beta_N - \varepsilon_N)\ell f_0 > 0 \tag{A.36}$$

for every $N \ge 40$. The finitely many orders $3 \le N \le 39$ are solved exactly over $\mathbb{Q}$ by Lemma 15; the first values are $\mathcal{S}_3 = 3/208$ and $\mathcal{S}_4 = 359/26660$.

The supplementary verifier reconstructs every rational boundary solve, cleared polynomial, shifted coefficient list, and the labelled normalizations (A.12s) and (A.17). Combining the three subsections proves Theorem 9.

B Two symmetric weighted K_4 families

Let $G_{13}(x)$ have one core vertex, three satellites, unit core–satellite weights, and satellite–satellite weight $x > 0$. Let $G_{22}(x, y)$ have two vertex pairs, internal-pair weights $x, y > 0$, and four unit cross-pair weights. These are the full $S_1 \times S_3$ and $S_2 \times S_2$ invariant families after common scaling.

$G_{13}(x)$ $G_{22}(x, y)$

Figure 2: The two invariant weighted K_4 families. Unlabeled edges have unit weight.

A two-class graph with class sizes p, q, internal weights α, β, and cross weight γ is strongly lumpable by mutant counts (i, j). The four state-changing probabilities follow directly from (2.2):

$$\begin{aligned}
P_{i,j}^{i+1,j} &= \frac{p-i}{p+q}\frac{r(i\alpha+j\gamma)}{r(i\alpha+j\gamma)+(p-i-1)\alpha+(q-j)\gamma},\\
P_{i,j}^{i-1,j} &= \frac{i}{p+q}\frac{(p-i)\alpha+(q-j)\gamma}{r((i-1)\alpha+j\gamma)+(p-i)\alpha+(q-j)\gamma},\\
P_{i,j}^{i,j+1} &= \frac{q-j}{p+q}\frac{r(j\beta+i\gamma)}{r(j\beta+i\gamma)+(q-j-1)\beta+(p-i)\gamma},\\
P_{i,j}^{i,j-1} &= \frac{j}{p+q}\frac{(q-j)\beta+(p-i)\gamma}{r((j-1)\beta+i\gamma)+(q-j)\beta+(p-i)\gamma}.
\end{aligned} \tag{B.1}$$

For the $1+3$ family, exact elimination gives

$$\rho_{\mathrm{dB}}(G_{13}(x), r) - \rho_{\mathrm{dB}}(J_4, r) = -\frac{3r^2(r-1)(x-1)^2 F_{13}}{4(r^2+r+1)P_{13}}, \tag{B.2}$$

where

$$\begin{aligned}
F_{13} = {}& 2r^4x^2 + 2r^4x + 11r^3x^2 + 14r^3x + 3r^3 + 21r^2x^2 + 29r^2x + 12r^2\\
&+ 16rx^2 + 22rx + 8r + 4x^2 + 5x + 1,
\end{aligned}$$

and

$$\begin{aligned}
P_{13} = {}& 8x^2(x+1)(r^6+1) + (6x^4 + 36x^3 + 46x^2 + 16x)(r^5 + r)\\
&+ (27x^4 + 73x^3 + 85x^2 + 59x + 12)(r^4 + r^2)\\
&+ (42x^4 + 90x^3 + 106x^2 + 66x + 24)r^3.
\end{aligned}$$

Every coefficient is positive.

For the $2+2$ family,

$$\rho_{\mathrm{dB}}(G_{22}(x, y), r) - \rho_{\mathrm{dB}}(J_4, r) = -\frac{r^2(r-1)H_{22}(x, y, r)}{4(r^2+r+1)P_{22}(x, y, r)}. \tag{B.3}$$

The reduced P_{22} has 123 positive integer monomials. Independently, the holding-step-free transient matrix M_{22} satisfies

$$\det M_{22} = \frac{P_{22}}{128 L_{22}}, \tag{B.4}$$

where

$$L_{22} = (2r+x)(2r+y)(rx+2)(ry+2)(r+x+1)(r+y+1)$$
$$\times (rx+r+1)(ry+r+1) > 0.$$

Absorption makes M_{22} a nonsingular M-matrix, so $P_{22} > 0$.

Set $g = \sqrt{xy} > 0$, $d = (\sqrt{x} - \sqrt{y})^2 \geq 0$, and $t = r - 1 > 0$. Then

$$H_{22} = C_0(g,t) + C_1(g,t)d + C_2(g,t)d^2 + C_3(g,t)d^3 + C_4(g,t)d^4. \tag{B.5}$$

Here

$$C_0 = 2t(g-1)^2(g+1)(t+1)R_0,$$

$$\begin{aligned} R_0 = {} & (2g^2+2g)t^4 + (g^3+10g^2+21g+6)t^3 \\ & + (3g^3+26g^2+61g+38)t^2 + (4g^3+32g^2+80g+64)t \\ & + 2g^3+16g^2+40g+32, \end{aligned}$$

and

$$\begin{aligned} C_1 = {} & 2(g^4+4g^3-2g^2+4g+1)t^6 + (11g^4+108g^3+76g^2+60g+17)t^5 \\ & + 2(40g^4+288g^3+393g^2+176g+19)t^4 \\ & + (16g^5+331g^4+1704g^3+2766g^2+1360g+39)t^3 \\ & + 2(28g^5+337g^4+1426g^3+2399g^2+1378g+12)t^2 \\ & + 2(g+2)(32g^4+263g^3+722g^2+661g+2)t \\ & + 24g(g+2)(g+4)(g^2+4g+5), \\ C_2 = {} & 2(g^2+1)t^6 + 3(9g^2+26g+5)t^5 + 2(18g^3+120g^2+321g+44)t^4 \\ & + 2(2g^4+105g^3+525g^2+1071g+170)t^3 \\ & + (14g^4+474g^3+2201g^2+3648g+689)t^2 \\ & + 2(8g^4+240g^3+1080g^2+1587g+331)t \\ & + 6(g^4+30g^3+133g^2+186g+40), \\ C_3 = {} & 13t^5 + (6g^2+48g+107)t^4 + (35g^2+312g+357)t^3 \\ & + (79g^2+744g+608)t^2 + (80g^2+768g+529)t + 6(5g^2+48g+31), \\ C_4 = {} & 6t^4+39t^3+93t^2+96t+36. \end{aligned}$$

Every displayed coefficient is positive except for the single grouped factor

$$g^4+4g^3-2g^2+4g+1 = (g^2-1)^2 + 4g(g^2+1) > 0. \tag{B.6}$$

Thus $C_j > 0$ for $j \geq 1$ and $C_0 \geq 0$. If $d > 0$, then $C_1 d > 0$; if $d = 0$, $H_{22} = C_0$ vanishes only at $g = 1$.

Theorem 16 (symmetric weighted K_4 families)**.** *For every $r > 1$,*

$$\rho_{\mathrm{dB}}(G_{13}(x), r) \leq \rho_{\mathrm{dB}}(J_4, r), \qquad \rho_{\mathrm{dB}}(G_{22}(x,y), r) \leq \rho_{\mathrm{dB}}(J_4, r).$$

Equality holds only at $x = 1$ in the first family and $x = y = 1$ in the second.

This is not a classification of the unrestricted six-edge weighted K_4.

C Quantifier and scope ledger

The statements in this paper have deliberately different orders of quantification.

Fitness-two local theorem. For every $n \geq 3$ there is an $\eta_n > 0$ such that every positive loopless row-stochastic P with $0 < \|P - J_n\|_F < \eta_n$ satisfies $\rho_{\mathrm{dB}}(P, 2) < \rho_{\mathrm{dB}}(J_n, 2)$.

Full-tangent curvature. For every $n \geq 3$ and every $\delta \in \mathcal{T}_n \setminus \{0\}$,

$$D\rho_{\mathrm{dB}}(J_n, 2)[\delta] = 0, \qquad D^2\rho_{\mathrm{dB}}(J_n, 2)[\delta, \delta] < 0.$$

Fixed-graph strong selection. For every fixed normalized kernel $P \neq J_n$, there is an $r_0(P)$ such that $\rho_{\mathrm{dB}}(P, r) < \rho_{\mathrm{dB}}(J_n, r)$ for every $r > r_0(P)$.

Triangle theorem. Every positive weighted triangle and every $r > 1$ obey complete-graph maximality, with strictness away from equal weights.

Open global fitness-two statement. It remains open whether every n and every kernel P induced by an undirected weighting satisfies $\rho_{\mathrm{dB}}(P, 2) \leq \rho_{\mathrm{dB}}(J_n, 2)$.

Open asymptotic-family statement. It also remains open whether there is one sequence (P_n) such that, for every fixed $r > 1$, some $n_0(r)$ makes every P_n with $n \geq n_0(r)$ an amplifier.

The proved rows do not imply either open row. In particular, local optimality has a radius η_n that need not be bounded below, and the strong-selection threshold $r_0(P_n)$ may diverge with n.

References

B. Allen, G. Lippner, Y.-T. Chen, B. Fotouhi, N. Momeni, S.-T. Yau, and M. A. Nowak. Evolutionary dynamics on any population structure. *Nature* **544** (2017), 227–230. doi:10.1038/nature21723.

B. Allen, C. Sample, R. Jencks, J. Withers, P. Steinhagen, L. Brizuela, J. Kolodny, D. Parke, G. Lippner, and Y. A. Dementieva. Transient amplifiers of selection and reducers of fixation for death–Birth updating on graphs. *PLoS Computational Biology* **16** (2020), e1007529. doi:10.1371/journal.pcbi.1007529.

A. M. Etheridge and R. C. Griffiths. A coalescent dual process in a Moran model with genic selection. *Theoretical Population Biology* **75** (2009), 320–330. doi:10.1016/j.tpb.2009.03.004.

J. Foo, E. B. Gunnarsson, K. Leder, and D. Sivakoff. Dynamics of advantageous mutant spread in spatial death–birth and birth–death Moran models. *Communications on Applied Mathematics and Computation* **6** (2024), 576–604. doi:10.1007/s42967-023-00278-6.

T. E. Harris. Additive set-valued Markov processes and graphical methods. *The Annals of Probability* **6** (1978), 355–378. doi:10.1214/aop/1176995523.

L. Hindersin and A. Traulsen. Most undirected random graphs are amplifiers of selection for birth–death dynamics, but suppressors of selection for death–birth dynamics. *PLoS Computational Biology* **11** (2015), e1004437. doi:10.1371/journal.pcbi.1004437.

K. Kaveh, N. L. Komarova, and M. Kohandel. The duality of spatial death–birth and birth–death processes and limitations of the isothermal theorem. *Royal Society Open Science* **2** (2015), 140465. doi:10.1098/rsos.140465.

A. Kriebel. No universal death–Birth amplifier on a finite weighted population structure. Unrefereed predecessor manuscript in the version 1.0.0 source-and-software release (2026). doi:10.5281/zenodo.21753405.

A. Kriebel. Simultaneous amplification beyond fitness three halves. Unrefereed companion manuscript in the version 1.0.0 source-and-software release (2026). doi:10.5281/zenodo.21852072.

S. M. Krone and C. Neuhauser. Ancestral processes with selection. *Theoretical Population Biology* **51** (1997), 210–237. doi:10.1006/tpbi.1997.1299.

E. Lieberman, C. Hauert, and M. A. Nowak. Evolutionary dynamics on graphs. *Nature* **433** (2005), 312–316. doi:10.1038/nature03204.

S. Mano. Duality, ancestral and diffusion processes in models with selection. *Theoretical Population Biology* **75** (2009), 164–175. doi:10.1016/j.tpb.2009.01.007.

N. Masuda and H. Ohtsuki. Evolutionary dynamics and fixation probabilities in directed networks. *New Journal of Physics* **11** (2009), 033012. doi:10.1088/1367-2630/11/3/033012.

P. A. P. Moran. Random processes in genetics. *Mathematical Proceedings of the Cambridge Philosophical Society* **54** (1958), 60–71. doi:10.1017/S0305004100033193.

H. Richter. Spectral dynamics of guided edge removals and identifying transient amplifiers for death–Birth updating. *Journal of Mathematical Biology* **87** (2023), article 3. doi:10.1007/s00285-023-01937-1.

P. Shakarian, P. Roos, and A. Johnson. A review of evolutionary graph theory with applications to game theory. *BioSystems* **107** (2012), 66–80. doi:10.1016/j.biosystems.2011.09.006.

J. Svoboda, S. Joshi, J. Tkadlec, and K. Chatterjee. Amplifiers of selection for the Moran process with both Birth–death and death–Birth updating. *PLoS Computational Biology* **20** (2024), e1012008. doi:10.1371/journal.pcbi.1012008.

J. Tkadlec, A. Pavlogiannis, K. Chatterjee, and M. A. Nowak. Limits on amplifiers of natural selection under death–Birth updating. *PLoS Computational Biology* **16** (2020), e1007494. doi:10.1371/journal.pcbi.1007494.